

Literature (High) Results

Objective

Edison Literature — three queries for E1 (negotiation testbed)

Run them separately, in this order. Each is self-contained. L1 and L2 answer "how do I measure uncertainty" and "how do I exploit it instead of getting conservative"; L3 answers "which credit mechanism should the coordination layer use". Do not merge them: Literature is a synthesis agent and performs better on one bounded question at a time.

Answer format requested from every query: a synthesis organized by the sub-questions below, not a paper-by-paper list; each claim cited; a closing section listing disagreements in the literature, findings that failed to replicate, and questions the literature does not answer.

L1 — Measuring uncertainty about a counterpart's hidden parameters

****Query.****

In automated bilateral multi-issue negotiation, an agent maintains a belief over the counterpart's hidden parameters: reservation value, issue weights, deadline, and concession profile. I need a synthesis of how that belief is represented, updated, and evaluated as a probabilistic object rather than a point estimate.

Cover:

1. Representations of the belief over the reservation value specifically: Bayesian learning from the concession sequence, non-linear regression on the opponent's offer curve, particle filters or Kalman-type filters, and Gaussian process or Bayesian neural approaches. What is assumed about stationarity in each?
2. Evaluation. Is opponent-model accuracy in this field reported with proper scoring rules (log score, Brier, CRPS) and calibration diagnostics, or only with point-error measures such as RMSE on the estimated reservation value and utility distance? Where proper scoring or calibration has been used, on what benchmarks, and what was found?
3. Decomposition of the uncertainty: is the distinction between irreducible (aleatoric) and reducible (epistemic) uncertainty made explicit in opponent modeling, and does

any work estimate the value of perfect information about the counterpart's reservation value?

4. How the belief's uncertainty evolves within a session: does the posterior actually concentrate as offers accumulate, at what rate, and under which opponent families does it fail to concentrate or converge to the wrong value?
5. Which parameters are learnable at all. Reported evidence on the identifiability of reservation value versus deadline versus concession rate from an offer sequence, including any impossibility or confounding results.

Anchor points to start from and expand beyond: the BOA architecture decomposing negotiating agents into bidding strategy, opponent model, and acceptance strategy; Baarslag and colleagues' survey and comparative evaluation of opponent models; Zeng and Sycara's Bazaar; Faratin, Sierra and Jennings on negotiation tactics; the ANAC competition series with the GENIUS and NegMAS platforms.

Exclude: unstructured natural-language dialogue negotiation with no explicit issue space, and pure game-theoretic equilibrium analysis with no learning component.

L2 — Exploiting uncertainty instead of conceding to it

****Query.****

An agent that maximizes expected utility under a wide posterior over the counterpart's reservation value produces conservative offers, because the estimated probability of acceptance falls quickly away from the posterior mean. I need a synthesis of mechanisms that turn uncertainty into an asset rather than a reason to concede.

Cover:

1. Offers as experiments. Work that treats a proposal as an information-gathering action and selects it by expected information gain, Bayesian experimental design, or active-learning criteria, and any explicit treatment of the exploration-exploitation tradeoff inside a single negotiation session with a finite deadline.
2. Optimism under uncertainty. Decision rules based on an upper quantile of the belief rather than its mean (UCB-style, distributionally optimistic, or risk-seeking utilities), applied to bargaining or to sequential pricing and posted-price problems that share the structure.
3. Integrative mechanisms that require uncertainty or disagreement to create value: logrolling across issues with differing priorities, multiple equivalent simultaneous offers as a probing device, and contingent contracts that let parties bet on differing beliefs rather than resolve them.
4. The cost side. Empirical relationships between offer aggressiveness or ambitious first offers and impasse rate, walkaway, and relationship damage; whether a

utility-versus-impasse frontier has been characterized quantitatively anywhere, and how the optimum shifts with the number of rounds remaining.

5. Any result showing that a bolder policy is better precisely because it is more informative, separating the information effect from the anchoring effect.

Anchor points: anchoring and first-offer effects in the negotiation literature; Bazerman and Gillespie on contingent contracts; MESOs; Lindley's expected information gain; Bayesian optimization acquisition functions applied to sequential decisions with a deadline.

Include both computational and behavioral evidence, and mark clearly which is which.

L3 — Credit assignment in ensembles of sequential probabilistic predictors

****Query.****

I have a set of heterogeneous models that each produce a posterior over the same hidden parameter and update sequentially as evidence arrives. Ground truth is revealed only at the end of each episode, with partial signals in between. I need a synthesis of the mechanisms available to attribute the ensemble's performance back to individual models, and evidence on which ones preserve ensemble diversity over repeated rounds of reweighting.

Cover:

1. The mechanism families and their formal guarantees: Shapley value attribution and its sampling approximations; difference rewards and counterfactual baselines from cooperative multi-agent learning; Hanson's market scoring rules and the logarithmic market scoring rule as a sequentially-shared belief with per-contributor payoff; exponential weights and prediction with expert advice; Bayesian model averaging and stacking. What each guarantees and what each assumes.
2. Equivalences and differences among them. Where difference rewards coincide with Shapley value, and where market scoring payoffs diverge from either.
3. Diversity under selection. The ambiguity or diversity decomposition of ensemble error and empirical evidence on whether repeated reweighting or pruning by individual accuracy collapses diversity and degrades the ensemble; any mechanism reported to preserve diversity while still reweighting.
4. Incentive pathologies in repeated settings: herding and information cascades in market scoring rules, free-riding under shared reward, and reported mitigations.
5. Empirical comparisons in forecasting ensembles with resolved outcomes, especially collaborative forecast hubs: how often does a credit-weighted or learned-weight combination beat the equal-weight ensemble, by how much, and under what conditions does the equal-weight baseline prove hard to beat.

6. Credit assignment with delayed and partial feedback: temporal credit assignment, eligibility traces, and any application of these to ensembles of forecasters rather than to policy learning.

Anchor points: Shapley; Wolpert and Tumer on difference rewards; Hanson on market scoring rules; Krogh and Vedelsby on the ambiguity decomposition; Cesa-Bianchi and Lugosi on prediction with expert advice; the COVID-19 and influenza forecast hubs.

Where results are domain-specific, say so; I need to know what transfers to a setting with roughly six models, a few thousand episodes, and a low-dimensional hidden parameter.

Results

L1 — Measuring Uncertainty About a Counterpart's Hidden Parameters

1. Representations of the Belief Over the Reservation Value

The literature on automated bilateral negotiation employs several distinct representations for an agent's belief about the counterpart's hidden parameters. The earliest and most widely adopted approach is **Bayesian hypothesis-space learning**, pioneered by Zeng and Sycara 1998. In this framework, the agent maintains a discrete set of hypotheses—each assigning a candidate value to the opponent's reservation price—together with prior probabilities. Observed offers and counter-offers serve as signals; the hypothesis probabilities are updated via Bayes' rule, and the estimate is computed as a likelihood-weighted sum of candidate values ([1.1], [1.2]). This approach implicitly assumes that the opponent's reservation value is fixed (stationary) within a single negotiation session, though the Bazaar system was designed for an "open-world" setting in which environment parameters could change ([1.1]).

A second family uses **nonlinear regression on the opponent's offer curve**. Here, the reservation value is treated as a parameter of a fitted decision function—typically the Faratin–Sierra–Jennings time-dependent tactic, parameterized by reservation value, deadline, and a concession-shape exponent β ([2.1], [2.2]). Regression assumes a known functional form for the tactic (e.g., polynomial or exponential concession) and estimates its parameters from observed bids. This approach assumes stationarity in the sense that the opponent's tactic type and parameters do not change mid-session ([3.1], [3.2]).

Particle filters have been applied to opponent modeling in multi-criteria negotiation. Eshragh 2019 used unscented particle filtering to recursively estimate the parameters of fuzzy preference models from feedback received at each negotiation round. The prediction–correction cycle of Bayesian filtering allows progressive refinement as offers accumulate, though no formal convergence guarantee was established ([4.1], [4.2]). The stationarity assumption is that opponents' strategies do not change during negotiation.

Gaussian process (GP) models were introduced by Leahu, Kaisers, and Baarslag 2019, who proposed a stochastic utility model in which both the user's and opponent's utility functions are represented as random functions under a GP prior. The posterior is updated via Laplace approximation as interaction data

(comparative queries, offers) accumulate. The GP framework naturally represents uncertainty as a full posterior distribution rather than a point estimate ([5.1], [5.2]). Stationarity is assumed in that the underlying utility function is fixed; the GP captures uncertainty about its shape rather than temporal drift.

More recently, **LLM-augmented Bayesian models** have extended discrete hypothesis updating by incorporating qualitative signals from natural-language utterances, using Luce's Choice Axiom for linguistic likelihoods and normal-distribution likelihoods for numerical offers, fused through Bayesian updating ([6.1], [6.2]).

2. Evaluation: Scoring Rules and Calibration

The field's evaluation of opponent models overwhelmingly relies on **point-error and outcome-based measures** rather than proper scoring rules. The comprehensive survey by Baarslag et al. 2016 catalogues the following accuracy metrics: Euclidean distance on issue weights, mean squared error and mean absolute error on predicted bid utilities, Pearson correlation for preference rankings, and percentage error for reservation-value and deadline estimates ([3.3], [3.4]). On the performance side, models are evaluated by negotiation-outcome metrics: average utility, distance to the Pareto frontier, distance to a fair solution (Nash or Kalai–Smorodinsky), agreement percentage, and joint utility ([3.5], [3.6]). The survey explicitly notes that accuracy metrics for reservation-value and deadline estimators remain underdeveloped ([3.7]).

Proper scoring rules and calibration diagnostics are **almost entirely absent** from the core automated-negotiation literature. Two notable exceptions come from adjacent work. Cui and Lin 2025 introduced BOND (Bayesian Opponent-belief Negotiation Distillation), which reports turn-level **Brier scores** for opponent-type prediction in the CaSiNo negotiation domain. The Bayesian teacher achieved a mean Brier score of 0.085 on 1,054 predictions, compared to 0.114 for the distilled student and 0.194 for a chain-of-thought baseline—the first use of a proper scoring rule for opponent beliefs in negotiation ([7.1], [7.2], [7.3]). Sicilia et al. 2024 proposed FortUne Dial, evaluating negotiation-outcome forecasting across eight corpora using Brier score and Brier skill score, with explicit calibration analysis ([8.1], [8.2]). These works demonstrate that proper scoring rules are feasible and informative, but they remain isolated contributions.

3. Decomposition of Uncertainty: Aleatoric vs. Epistemic

The aleatoric–epistemic distinction is **not made explicit** in the mainstream automated-negotiation opponent-modeling literature. The Baarslag et al. 2016 survey frames uncertainty purely as incomplete knowledge of the opponent's private preferences and strategy—what would formally be epistemic uncertainty—without separating it from irreducible randomness in opponent behavior ([3.7], [3.8]).

The distinction has been formalized in two adjacent lines of work. **Adversarial Risk Analysis** (Banks et al., 2022) explicitly separates aleatory uncertainty (randomness in outcomes conditional on both parties' choices) from epistemic uncertainty (incomplete knowledge of the opponent's utility function, beliefs, and resources). ARA models the latter through subjective probability distributions over the opponent's attributes and uses Monte Carlo simulation to propagate these uncertainties ([9.1], [9.2]). **UAOM** (Yang et al., 2025) provides an explicit aleatoric–epistemic decomposition in deep RL opponent modeling, using a hybrid of deterministic and probabilistic encoders: the probabilistic component captures behavioral randomness (aleatoric), while the ensemble of models captures knowledge limitations (epistemic) ([10.1], [10.2], [10.3]). No work was found that estimates the **value of perfect information** specifically about the counterpart's reservation value, though Hendrikx 2012 benchmarks performance under perfect knowledge of the full preference profile as an upper bound ([11.1], [12.1]).

4. Posterior Concentration Within a Session

Multiple papers describe Bayesian updating that should in principle narrow the posterior as offers accumulate, but **no formal convergence rate or posterior-concentration result** has been established in the negotiation literature. Hindriks and Tykhonov 2008 describe a spread parameter σ controlling how aggressively the posterior updates: lower σ reflects greater prior confidence and produces faster learning, while higher σ reflects uncertainty and makes updating less aggressive ([13.1]). Eshragh's 2019 particle-filter approach demonstrates improved opponent estimates over rounds, but without formal convergence guarantees ([14.1], [4.3]). The GP model of Leahu et al. 2019 updates beliefs as interaction data accumulate, but does not establish convergence conditions ([5.1]).

The literature implicitly acknowledges that concentration **fails** against certain opponent families. When the opponent uses a Boulware strategy ($\beta < 1$), concessions are withheld until near the deadline, providing minimal information for most of the session ([2.1]). Against behavior-dependent or strategic opponents who condition their offers on the agent's behavior, the stationarity assumption underlying most models is violated, and the posterior may converge to the wrong value ([13.1]).

5. Which Parameters Are Learnable: Identifiability

The literature reveals significant **identifiability concerns** but no formal impossibility results. Baarslag et al. 2016 note that existing reservation-value and deadline estimators are narrowly scoped, typically assuming that the form of the opponent's decision function is known while only its parameters are unknown; extending these methods to arbitrary strategies and initially unknown preferences is described as an open problem ([3.7]). Under some models, reservation value and deadline are structurally related—the opponent concedes toward reservation value as the deadline approaches—so one may be derivable from the other rather than independently identified ([3.9]). Zhang et al. 2025 note that observed offers may reflect both the underlying reserve and behavioral parameters such as concession willingness, and that a fully developed Bayesian dynamic model would be needed to disentangle them ([15.1]). No impossibility theorem or formal non-identifiability proof for the joint recovery of reservation value, deadline, and concession rate from offer sequences was found.

Disagreements, Failures to Replicate, and Open Questions

Disagreements. Whether opponent modeling improves negotiation outcomes at all is contested: Hendriks 2012 and Baarslag et al. 2016 document cases where a poor model performs worse than using no model, and the computational cost of modeling may reduce the number of bids explored ([12.2], [12.3]). Erez et al. 2017 argue that "playing the domain" rather than modeling the opponent can be competitive ([16.1]).

Failures to replicate. No systematic replication failures were identified, but the survey notes that results are highly setting-dependent: model performance varies dramatically with opponent type, domain, and protocol, and direct comparison across studies is difficult due to lack of common benchmarks ([3.5], [3.6]).

Open questions. (i) Whether proper scoring rules and calibration diagnostics can replace or complement point-error metrics as standard evaluation in negotiation. (ii) Whether the aleatoric–epistemic decomposition can be operationalized to estimate the value of information about specific hidden parameters. (iii) Whether formal identifiability results (or impossibility results) can be derived for the joint recovery of reservation value, deadline, and concession rate. (iv) Whether posterior concentration rates can be established for different opponent families. (v) How belief representations should adapt to non-stationary opponents who change strategy mid-session.

L2 — Exploiting Uncertainty Instead of Conceding to It

1. Offers as Experiments

The idea that a negotiation proposal can serve as an information-gathering action has been explored computationally and behaviorally. Liu et al. 2026 model concurrent multi-buyer bargaining as a finite-horizon exploration–exploitation problem: a seller with a limited message budget must decide whether to use communication turns for diagnostic probing (exploration) to discover hidden buyer valuations, or to concentrate on a promising buyer (exploitation). Their RLVR-trained seller learns strategic probing and price anchoring as mechanisms for gathering information while influencing counterparties' offers, substantially outperforming frontier LLMs that fixate on the current highest bid ([17.1], [17.2], [17.3]).

Behavioral evidence for offers-as-experiments comes from Ding and Lehrer 2026, who study bilateral bargaining with gradual information revelation. Buyers can use strategically intermediate offers as screening devices: acceptance identifies a low-value seller (with risk of loss), while rejection accelerates belief updating. The theoretical model predicts extensive experimentation, but laboratory subjects experiment far less than predicted, with fairness preferences substantially reducing probing behavior ([18.1], [18.2]). This gap between theoretically optimal and observed experimentation is a key finding.

The GP-based negotiation model of Leahu et al. 2019 frames preference elicitation as an action with an expected information gain that must exceed its "bother" cost, providing a value-of-information interpretation for information-gathering actions during negotiation ([5.3]).

2. Optimism Under Uncertainty

No work was found that directly applies UCB-style or distributionally optimistic decision rules to bilateral negotiation in the traditional multi-issue setting. However, structural analogues exist in sequential pricing. The UCB and Thompson Sampling algorithms possess natural robustness properties for exploration under uncertainty in sequential decision problems with auxiliary information (Gur and Momeni, 2022). In nonstationary RL settings, Cheung et al. 2023 introduce "confidence widening"—adding additional optimism beyond the conventional UCB exploration bonus—to handle environments where historical data may be misleading, achieving improved dynamic regret bounds. These frameworks share the structure of negotiation (sequential decisions under uncertainty about a counterpart's hidden value) but have not been formally applied to it.

3. Integrative Mechanisms That Require Uncertainty or Disagreement

Multiple Equivalent Simultaneous Offers (MESOs) serve as a probing device: by presenting alternative packages with equivalent value to the proposer, the negotiator may obtain information about the counterpart's preferences while preserving flexibility. Park et al. 2019 implemented a MESO-based agent negotiating with humans over three issues and found that MESOs increased integrative settlements, settlement ratios, offer persuasiveness, and counterpart satisfaction relative to single-option offers ([19.1], [19.2]). The eAgora system similarly generates five potential offers close to a desired concession level, allowing selection that may accommodate differing preferences ([20.1]).

Logrolling across issues with differing priorities is the canonical integrative mechanism. Vetschera 2013 reports that a support tool recommending logrolling led users to reach agreements with higher joint utilities ([20.1]). Tomlinson and Lewicki 2015 recommend offering packages of equivalent self-value to diagnose the counterpart's priorities without full information disclosure ([21.1]). **Contingent contracts** are

mentioned as a mechanism for creating value from disagreement, but detailed computational treatments or empirical quantifications of their value were not found in the retrieved literature beyond brief mentions.

4. The Cost Side: Aggressiveness, Impasse, and the Utility-vs-Impasse Frontier

The most comprehensive quantitative characterization comes from Schweinsberg et al. 2023, who analyzed over 26 million eBay buyer–seller negotiations. The anchoring effect of first offers on final sale price is **strictly linear** across the full range of offer magnitudes ($R^2 = .997$, slope = 0.82): more ambitious offers produce proportionally lower prices conditional on agreement ([22.1]). In contrast, the relationship between offer aggressiveness and impasse risk is **nonlinear**, described by a quartic function with three zones: a safety zone with relatively low risk, an acceleration zone (approximately 90% to 20% of list price) with steadily increasing impasse risk, and a saturation zone below 20% where roughly 95% of negotiations end in impasse regardless of further aggressiveness ([22.2], [22.3]). The quantitatively estimated optimal buyer opening offer integrating both objectives is approximately **80% of list price**, though category-specific recommendations range from 33% to 95% ([22.4], [22.5]). Salient offer values (50%, 67%, 75%, 80%) serve as local risk minima, possibly reflecting cognitive salience of round fractions ([22.6]). This constitutes the clearest empirical characterization of a utility-versus-impasse frontier in the literature.

Lipp et al. 2023 extend this by introducing the "anchor zone" (distance between first offer and counteroffer) as a predictor of outcomes, finding that more aggressive counteroffers reduce subjective value for both parties ([23.1]).

5. Information Effect vs. Anchoring Effect

The Schweinsberg et al. 2023 findings are primarily interpreted through the anchoring lens rather than an information-gathering lens. The linear anchoring effect on price persists independently of impasse risk, suggesting it operates through cognitive adjustment rather than information revelation ([22.1]). Ames and Mason 2015 distinguish informational and politeness effects of range offers, finding that range offers can signal the offer-maker's reservation price and influence counterpart judgments of rudeness and reservation price ([23.1]). However, no work was found that cleanly **separates** the information effect (a bolder offer reveals more about the zone of possible agreement) from the anchoring effect (a bolder offer biases the counterpart's judgment) in a way that demonstrates that a bolder policy is better **precisely because** it is more informative. Ding and Lehrer's 2026 experimental evidence on screening offers comes closest, showing that intermediate offers function as costly experiments that accelerate belief updating, but this operates through seller acceptance/rejection rather than anchoring ([18.1]).

Disagreements, Failures to Replicate, and Open Questions

Disagreements. The behavioral literature documents that human negotiators under-explore relative to theoretical predictions ([18.2]), raising questions about whether optimal computational policies are implementable against human counterparts. Whether MESOs function primarily as probing devices or as persuasion mechanisms remains debated ([19.2]).

Open questions. (i) No work applies formal Bayesian experimental design or information-gain criteria to select offers within a single bilateral negotiation session with a finite deadline. (ii) UCB-style optimism has not been formally applied to bilateral bargaining. (iii) The clean separation of information and anchoring effects remains unaccomplished. (iv) Contingent contracts as a computational mechanism for

automated agents exploiting differing beliefs lack formal treatment. (v) How the optimal exploration–exploitation balance shifts with the number of rounds remaining has not been characterized for negotiation specifically.

L3 — Credit Assignment in Ensembles of Sequential Probabilistic Predictors

1. Mechanism Families and Formal Guarantees

Shapley value attribution assigns each model its average marginal contribution across all possible coalitions. It uniquely satisfies efficiency (contributions sum to total), symmetry, linearity, and null-player properties. Exact computation requires evaluating 2^N coalitions, making it exponential in the number of models; sampling approximations (Random Order Value, permutation sampling) reduce this to polynomial cost with probabilistic guarantees ([22.4]). Li et al. 2021 apply Shapley values to multi-agent credit assignment in cooperative RL, using Monte Carlo sampling to make computation tractable.

Difference rewards and counterfactual baselines (Wolpert and Tumer, 2002) assign each agent the difference between the global reward and a counterfactual reward computed by replacing that agent's action with a default. COMA (Foerster et al., 2018) implements this via a centralized critic that marginalizes out a single agent's action while keeping others fixed. Difference rewards assume access to a simulator or model to compute the counterfactual and guarantee that the individual reward is aligned with the global objective when the default action is appropriately chosen.

Hanson's market scoring rules and the **Logarithmic Market Scoring Rule (LMSR)** provide a mechanism where sequential contributors to a shared probabilistic belief receive payoffs equal to the improvement in the scoring rule they produce. The LMSR is sequentially shared: each contributor's payoff depends only on the state of the belief before and after their update, and the total payout is bounded by the liquidity parameter. The principal's worst-case loss is finite and known ex ante. Market scoring rules incentivize truthful reporting under risk neutrality.

Exponential weights / prediction with expert advice (Cesa-Bianchi and Lugosi) maintains a weight vector over experts, multiplicatively updating weights based on observed losses. The regret bound is $O(\sqrt{T \ln N})$ for N experts over T rounds with bounded losses, and second-order bounds depending on the cumulative squared payoffs are available. No distributional assumptions on the data sequence are required—the guarantees are worst-case over arbitrary sequences.

Bayesian Model Averaging (BMA) assigns posterior probabilities to models based on marginal likelihoods and combines predictions as a posterior-weighted average. Under correct model specification, BMA is optimal for prediction. **Stacking** (Wolpert, 1992; Breiman, 1996) learns combination weights by minimizing cross-validated loss, making fewer assumptions than BMA about model correctness.

2. Equivalences and Differences

Difference rewards and Shapley values coincide when the coalition-value function is linear in the agents' contributions, because both reduce to the marginal contribution. For nonlinear interactions, Shapley values average over all orderings of coalition formation, while difference rewards compute a single counterfactual against a default. Li et al. 2021 combine Shapley structure with counterfactual baselines in "Shapley Counterfactual Credits," using Shapley's coalition-averaging to improve upon single-counterfactual difference rewards.

Market scoring rule payoffs diverge from both: they are path-dependent (the payoff depends on when the contributor acts in the sequence) and incentive-compatible only under risk neutrality. Under risk aversion, market scoring rules act as opinion pools rather than as individual-incentive mechanisms.

3. Diversity Under Selection

The **ambiguity decomposition** (Krogh and Vedelsby, 1995) shows that the mean squared error of a weighted ensemble decomposes as: ensemble error = mean individual error – ambiguity, where ambiguity measures the weighted variance of individual predictions around the ensemble prediction. This establishes that diversity (ambiguity) directly reduces ensemble error. Brown and Wyatt 2003 extended this decomposition and showed that Negative Correlation Learning can explicitly optimize the tradeoff between individual accuracy and diversity. Audhkhasi et al. 2013 proposed a Generalized Ambiguity Decomposition extending the result beyond squared loss.

Repeated reweighting or pruning by individual accuracy can collapse diversity. When poor-performing models are downweighted, the ensemble approaches the behavior of its best member, losing the ambiguity benefit. Lee et al. 2024 propose HyRe, which dynamically reweights ensemble members at test time using a small set of labeled examples, maintaining diversity through regularization. L2 regularization of ensemble weights has been shown to preserve broader weight distribution across models and improve performance in COVID-19 forecasting ensembles (Stephens et al., 2023).

4. Incentive Pathologies in Repeated Settings

Information cascades can arise in market scoring rules when later participants observe earlier contributions and rationally ignore their private information, leading to herding. **Free-riding** occurs under shared rewards when individual agents can receive team benefits without contributing, reducing the informativeness of individual signals. Difference rewards are specifically designed to mitigate free-riding by isolating each agent's contribution through counterfactual comparison. In repeated ensemble settings, the analog of free-riding is a model that copies the ensemble's previous prediction, contributing no new information while sharing in collective credit.

5. Empirical Comparisons in Forecast Hubs

The COVID-19 Forecast Hub evaluation (Cramer et al., 2022) found that a simple **equally-weighted median ensemble** performed better than 83% of participating models for incident cases (N=886 predictions from 23 models) and 91% for deaths (N=763 from 20 models) across 32 European countries over 52 weeks. Crucially, among several ensemble methods, "the most influential and best choice was to use a median average of models instead of using the mean, regardless of methods of weighting component forecast models." Retrospective exploration of performance-weighted alternatives did not consistently outperform the equal-weight median.

For influenza forecasting, McAndrew and Reich 2021 developed an **adaptive stacking ensemble** that updates weights week-by-week using regularized likelihood. It outperformed an equal-weighted ensemble and achieved similar performance to a static ensemble that required a full season of training data, using only a fraction of available data. Tsang et al. 2024 report that an **Adaptive Weight Blending Ensemble (AWBE)** for influenza in Hong Kong reduced RMSE by 52% and WIS by 53%, substantially outperforming both individual models and a simple average ensemble.

Stephens et al. 2023 found that for COVID-19 mortality, a performance-based ensemble with L2 regularization outperformed a baseline mean ensemble across US states on both absolute error and weighted interval scores, but emphasized that "sufficient regularization and constraints" are essential to prevent over-

fitting of weights—consistent with the diversity-preservation principle.

The emerging pattern: the equal-weight baseline is hard to beat in real-time forecasting with heterogeneous, time-varying model pools; learned weights can improve performance when (a) the training window is appropriately chosen, (b) regularization prevents weight collapse, and (c) the ensemble is moderately sized.

6. Credit Assignment with Delayed and Partial Feedback

Temporal credit assignment through eligibility traces ($TD(\lambda)$) is well-established in policy learning but has not been directly applied to ensembles of forecasters. The adaptive ensemble approaches in forecast hubs (McAndrew and Reich, 2021; Tsang et al., 2024) update weights based on recently resolved outcomes, implementing a form of temporal credit with recency weighting. The adaptive multi-model ensemble of Fiandrino et al. 2026 applies a particle-filtering approach to epidemic projections, dynamically selecting individual model trajectories based on observed data throughout the projection period—a form of sequential credit assignment with partial feedback.

For the target setting (six models, a few thousand episodes, low-dimensional hidden parameter), the most transferable findings are: (i) exponential weights with moderate regularization provides the strongest formal guarantees with minimal assumptions; (ii) Shapley-based attribution is computationally feasible at six models (only 64 coalitions) and provides the fairest decomposition; (iii) regularized stacking or adaptive weighting can outperform equal weights given sufficient episodes; (iv) diversity preservation through regularization is essential—aggressive pruning or reweighting degrades ensemble performance.

Disagreements, Failures to Replicate, and Open Questions

Disagreements. Whether learned weights reliably outperform equal weights is the central empirical controversy. COVID-19 Forecast Hub results suggest they often do not; influenza and other settings suggest they can with appropriate regularization and adaptation.

Open questions. (i) How to set the regularization strength for weight learning as a function of ensemble size and episode count. (ii) Whether market scoring rule incentives transfer to non-strategic model ensembles. (iii) How to apply eligibility traces to attribute delayed outcomes to interim probabilistic predictions in forecasting ensembles. (iv) Whether Shapley-based credit preserves diversity better than accuracy-based pruning in repeated rounds. (v) The interaction between diversity preservation and temporal adaptation—how aggressively to update weights in response to recent performance while maintaining ensemble breadth.

References

[1.1] Bayesian learning in negotiation. DAJUN ZENG, KATIA SYCARA. *Int. J. Hum. Comput. Stud.* (1998). <https://doi.org/10.1006/ijhc.1997.0164> Context: "The excerpt describes counterpart reservation value as a subjective probability distribution maintained by each player over relevant information entities. In the buyer–supplier example, the relevant entity is the supplier’s reservation price, which is unknown to the buyer and represented through hypotheses such as “RP_supplier = \$100.00” and “RP_supplier = \$130.00,” with prior probabilities. Offers and counter-offers constitute incoming signals, while domain knowledge is encoded as conditional statements linking observations to hypotheses. The buyer revises the reservation-price belief using the standard Bayesian rule, and Bayesian belief networks are used as the computational representation and updating mechanism. The text therefore supports Bayesian learning from observed offer sequences, rather than non-linear regression, particle filters, Kalman filters, Gaussian processes, or Bayesian neural approaches; those alternatives are not discussed in this excerpt. No explicit stationarity assumption is stated. Instead, the model is designed for an open-world setting in which relevant changes in the outside environment can affect later decisions, and it allows environment parameters to change over time. Thus, the excerpt suggests dynamically revisable beliefs rather than a fixed stationary counterpart model, but it does not specify formal stationarity conditions for any of the alternative methods."

[1.2] Bayesian learning in negotiation. DAJUN ZENG, KATIA SYCARA. *Int. J. Hum. Comput. Stud.* (1998). <https://doi.org/10.1006/ijhc.1997.0164> Context: "The excerpt frames automated bilateral negotiation as sequential decision making under uncertainty rather than as a static game-theoretic outcome calculation. Agents estimate the probable choices of the other player and update information during the negotiation. The authors explicitly distinguish their approach by modeling negotiation as a sequential decision-making task and using Bayesian updating as the underlying learning mechanism. The described interaction has Agent 1 selecting an offer using an offer-value function, after which Agent 2 evaluates whether to accept or counter-propose using a similar procedure. This supports representing beliefs about a counterpart through Bayesian learning from observed interaction or offer behavior, but the excerpt does not specify the detailed mathematical form of a reservation-value belief, nor does it describe concession-sequence likelihoods, nonlinear regression on offer curves, particle filters, Kalman filters, Gaussian processes, or Bayesian neural networks. Consequently, no stationarity assumption for those alternative representations can be inferred here. The passage instead emphasizes that practical negotiation involves computational complexity, bounded rationality, heuristic methods, and learning, and that the intended model should handle multi-agent learning and multi-issue or multi-criteria negotiation."

[2.1] Negotiation decision functions for autonomous agents. Peyman Faratin, Carles Sierra, Nick R. Jennings. *Robotics Auton. Syst.* (1998). [https://doi.org/10.1016/s0921-8890\(98\)00029-3](https://doi.org/10.1016/s0921-8890(98)00029-3) Context: "Faratin, Sierra, and Jennings model time-dependent tactics by making the next offer a function of elapsed time and the negotiation deadline (tmax). An agent’s initial offer is determined by a constant x_j within the issue’s value interval, while the offer function moves toward the issue’s reservation value as the deadline approaches. The concession function $q_j(t)$ is constrained to remain between 0 and 1, begin at x_j , and reach 1 at tmax. Two parameterized families are presented: polynomial and exponential functions, with β controlling the curve’s convexity and therefore the timing of concession. Boulware tactics use $\beta < 1$ and maintain the initial offer until time is nearly exhausted, then concede toward the reservation value. Conceder tactics use $\beta > 1$ and move rapidly toward the reservation value. The excerpt also distinguishes behavior-dependent tactics, which calculate the next offer from the opponent’s previous attitude or behavior; it notes that these tactics include different forms of imitation, although their detailed equations are outside the supplied passage. The broader model additionally treats time as a resource and extends the framework to dynamic deadlines and resource-estimation tactics."

[2.2] Negotiation decision functions for autonomous agents. Peyman Faratin, Carles Sierra, Nick R. Jennings. *Robotics Auton. Syst.* (1998). [https://doi.org/10.1016/s0921-8890\(98\)00029-3](https://doi.org/10.1016/s0921-8890(98)00029-3) Context: "Faratin, Sierra, and Jennings classify negotiation tactics according to the criterion used to generate issue values. Time-dependent tactics model concession as a function of the remaining time before a deadline: an agent is expected to concede more rapidly as the deadline approaches. The particular concession curve distinguishes tactics in this family, with the excerpt illustrating polynomial and exponential time functions using relative time t/t_{max} . Behavior-dependent, or imitative, tactics instead base a counteroffer on the opponent's previous behavior; they vary in what opponent behavior is imitated and how strongly. Such tactics are especially applicable when the agent is not under substantial time pressure and wants to avoid exploitation. The model allows multiple tactics to be combined for one issue through weighted averaging. For example, an agent can combine a time-dependent suggestion with an imitative suggestion, assigning larger weight to the tactic considered more important. The excerpt does not explicitly name the Boulware and Conceder strategies, nor does it provide their exact reservation-value/deadline parameterized equations, but it establishes the framework in which such strategies are represented by different time-dependent concession curves. Offers are evaluated against the agent's scoring function and current prospective counteroffer; an offer is accepted if it scores better, rejected at the deadline, or otherwise answered with a counteroffer."

[3.1] Learning about the opponent in automated bilateral negotiation: a comprehensive survey of opponent modeling techniques. Tim Baarslag, Mark J. C. Hendriks, Koen V. Hindriks, Catholijn M. Jonker. *Autonomous Agents and Multi-Agent Systems* (2016). <https://doi.org/10.1007/s10458-015-9309-1> Context: "The excerpt identifies two representations for the counterpart's reservation value. In Bayesian learning, the belief is represented as a discrete set of hypotheses, each assigning a possible value to the reservation value, together with prior probabilities. During negotiation, observed behavior—especially concession sequences—is used to update the hypothesis probabilities, and the estimate is the likelihood-weighted sum of candidate values. This approach presumes suitable domain knowledge for generating candidate values and priors; the related methods may additionally assume a particular time-dependent tactic and connect the opponent's reservation value with a private deadline. In non-linear-regression approaches, the reservation value is represented as a parameter of a fitted decision or offer function. Regression can estimate whether the opponent's tactic is time-, behavior-, or resource-dependent and infer its associated parameters, including the reservation value. The excerpt does not mention particle filters, Kalman filters, Gaussian processes, or Bayesian neural approaches. It also does not explicitly discuss stationarity assumptions. Instead, it emphasizes assumptions about known tactic forms, private or public deadlines, domain knowledge, and the opponent's concession behavior near the reservation value. Thus, the passage supports Bayesian hypothesis weighting and parametric curve fitting, but provides no evidence for the other listed representations or a detailed comparison of their stationarity requirements."

[3.2] Learning about the opponent in automated bilateral negotiation: a comprehensive survey of opponent modeling techniques. Tim Baarslag, Mark J. C. Hendriks, Koen V. Hindriks, Catholijn M. Jonker. *Autonomous Agents and Multi-Agent Systems* (2016). <https://doi.org/10.1007/s10458-015-9309-1> Context: "The excerpt discusses opponent-modeling methods relevant to time-dependent and behavior-dependent negotiation tactics. It identifies Boulware and Conceder as examples of time-dependent negotiation strategies. It also explains that an opponent's reservation value can be estimated from observed concessions, based on the assumption that the opponent stops conceding as its reservation value is approached, particularly near the negotiation deadline. In models involving private deadlines, the opponent's reservation value and deadline may be related so that one can be inferred from the other. A separate approach estimates whether the opponent uses a time-dependent, behavior-dependent, or resource-dependent tactic, while simultaneously estimating the tactic's parameters, including the reservation value. Non-linear re-

gression is one technique used for this purpose. The excerpt therefore supports discussion of concession functions whose behavior depends on parameters such as reservation value and deadline, and of methods for learning those parameters from offers and concessions. However, the specific Faratin–Sierra–Jennings model is not described by name in the supplied pages; only the broader categories of time-dependent and behavior-dependent tactics, together with Boulware and Conceder examples, are explicitly mentioned."

[3.3] Learning about the opponent in automated bilateral negotiation: a comprehensive survey of opponent modeling techniques. Tim Baarslag, Mark J. C. Hendriks, Koen V. Hindriks, Catholijn M. Jonker. *Autonomous Agents and Multi-Agent Systems* (2016). <https://doi.org/10.1007/s10458-015-9309-1> Context: "The excerpt describes two broad classes of evaluation: accuracy measures for the opponent model itself and performance measures for the negotiation strategy using that model. Reported accuracy metrics are primarily point-estimate or association measures. For issue-weight, reservation-value, or other scalar estimates, the survey gives Euclidean distance; it also reports rank disagreement and Pearson correlation for issue-weight vectors. For predicted bidding behavior, it reports mean squared error (MSE/RMSE-related), mean absolute error, percentage error, and correlation. For complete preference profiles, it reports mean absolute utility distance across outcomes, pairwise ranking distance, and Pearson correlation; these metrics can also evaluate acceptance-strategy models by comparing actual and predicted acceptance probabilities for bids. The excerpt notes scalability limitations for all-pairs ranking metrics and suggests Monte Carlo or correlation-based alternatives. It then states that strategy-level evaluation focuses on performance gains from employing the model, with negotiation duration mentioned as another possible measure. This passage contains no mention of proper scoring rules such as log score, Brier score, or CRPS, no calibration diagnostics, and no explicit benchmark protocol beyond the surveyed studies and metric overview tables. Thus, based on this excerpt, evaluation is centered on point-error, ranking, correlation, and downstream negotiation-performance measures."

[3.4] Learning about the opponent in automated bilateral negotiation: a comprehensive survey of opponent modeling techniques. Tim Baarslag, Mark J. C. Hendriks, Koen V. Hindriks, Catholijn M. Jonker. *Autonomous Agents and Multi-Agent Systems* (2016). <https://doi.org/10.1007/s10458-015-9309-1> Context: "The excerpt describes opponent-model evaluation as requiring both model-level accuracy measures and agent-level negotiation performance measures, embedded in a fair tournament with varied negotiation strategies and scenarios. The recommended accuracy metric depends on the model type: reservation-value and deadline estimators use percentage error, because signed over- or underestimation can affect agreement; acceptance-probability models use ranking distance or Pearson correlation over bids; issue-weight estimates use Euclidean distance; broader preference-profile estimates use ranked bid distance or Pearson correlation; and bidding-strategy predictions use mean squared error between actual and predicted utility, with greater emphasis on outliers than mean absolute error. The passage does not report proper probabilistic scoring rules such as log score, Brier score, or CRPS, nor does it discuss calibration diagnostics. Performance evaluation is tied to the agent's objective. Minimizing negotiation cost is assessed with time to agreement, percentage of agreements, and average utility. Adapting to the opponent is assessed with average utility, trajectory analysis, and potentially robustness. Win-win outcomes are associated in Table 5 with average utility, distance to the Pareto frontier, distance to a fair outcome, agreement percentage, and joint utility. The benchmark should use a large, balanced set of opponents and scenarios while avoiding experimental designs that favor particular modeling approaches."

[3.5] Learning about the opponent in automated bilateral negotiation: a comprehensive survey of opponent modeling techniques. Tim Baarslag, Mark J. C. Hendriks, Koen V. Hindriks, Catholijn M. Jonker. *Autonomous Agents and Multi-Agent Systems* (2016). <https://doi.org/10.1007/s10458-015-9309-1> Context: "The excerpt indicates that opponent models are evaluated using both negotiation-performance met-

rics and model-accuracy metrics, but it does not report a standardized benchmark or a systematic use of proper scoring rules such as log score, Brier score, or CRPS. For adaptation, the authors recommend average utility, trajectory analysis, and robustness analysis using evolutionary game theory, partly to ensure that adaptation does not lead to exploitation. For win-win outcomes, suggested measures include distance to the Pareto frontier, joint utility, and especially distance to a fair outcome. The survey emphasizes that existing studies generally evaluate models in their own negotiation settings, making direct comparison difficult, and explicitly calls for a common negotiation benchmark. For reservation-value and deadline estimation, the authors state that accuracy measures are still lacking; basic measures could be the distance between actual and estimated values, but more advanced methods remain undetermined. Acceptance-strategy accuracy is described as especially challenging, with no established measure. Thus, the excerpt supports point-distance/error-style evaluation and downstream utility/outcome metrics, while suggesting that calibration diagnostics and probabilistic scoring rules are not established or reported here. It also does not provide named benchmark datasets or a consolidated list of reported RMSE or utility-distance results."

[3.6] Learning about the opponent in automated bilateral negotiation: a comprehensive survey of opponent modeling techniques. Tim Baarslag, Mark J. C. Hendriks, Koen V. Hindriks, Catholijn M. Jonker. *Autonomous Agents and Multi-Agent Systems* (2016). <https://doi.org/10.1007/s10458-015-9309-1> Context: "This excerpt describes evaluation of opponent models primarily through their effect on negotiation performance, while also recommending separate accuracy measures. The dominant metric is average utility, commonly comparing agents with and without an opponent model across groups of opponents and multiple negotiation domains. Other reported metrics include distance to fair outcomes such as the Nash or Kalai-Smorodinsky solutions, distance to the Pareto frontier, joint utility, percentage of agreements, robustness, time or communication load, and trajectory analysis. The survey cautions that agreement percentage alone does not measure agreement quality and should be paired with average utility. Evaluation should use multiple opponents, domains, and preferably multiple negotiation strategies, since results depend strongly on whether the model's assumptions fit the negotiation setting. A large, balanced set of settings can reduce this bias. The excerpt explicitly recommends combining performance measures with accuracy measures from Section 6.1, but these pages do not identify log score, Brier score, CRPS, RMSE, reservation-value error, or utility-distance diagnostics. Thus, the passage provides evidence about outcome-level benchmarks and experimental design, but not enough detail to determine whether proper scoring rules or calibration diagnostics are used elsewhere in the survey."

[3.7] Learning about the opponent in automated bilateral negotiation: a comprehensive survey of opponent modeling techniques. Tim Baarslag, Mark J. C. Hendriks, Koen V. Hindriks, Catholijn M. Jonker. *Autonomous Agents and Multi-Agent Systems* (2016). <https://doi.org/10.1007/s10458-015-9309-1> Context: "The excerpt does not explicitly use the aleatoric-versus-epistemic uncertainty distinction, nor does it discuss the value of perfect information about an opponent's reservation value. It does, however, frame opponent modeling as learning either what the opponent wants—reservation value, deadline, and preference order—or what the opponent will do, including bidding and acceptance strategies. The survey emphasizes that behavior prediction is harder because it depends on the opponent's underlying preferences. Regarding identifiability, the excerpt indicates that existing reservation-value and deadline estimators are narrowly scoped: they concern single-issue quantitative negotiations, generally use Bayesian learning and/or nonlinear regression, and usually assume that the form of the opponent's decision function is known while only its parameters are unknown. The authors identify extending these methods to arbitrary strategies and initially unknown preferences as an open problem, describing it as a multilayered process involving preference learning and strategy prediction. For multi-issue preference estimation, linear-additive assumptions are common, while nonlinear interaction effects create scalability problems. The excerpt

does not provide formal confounding analyses or impossibility results showing whether reservation value, deadline, and concession rate are uniquely recoverable from an offer sequence. It instead notes that accuracy metrics for reservation value, deadline, and acceptance-strategy models remain underdeveloped; more generally, the lack of common benchmarks makes comparisons difficult."

[3.8] Learning about the opponent in automated bilateral negotiation: a comprehensive survey of opponent modeling techniques. Tim Baarslag, Mark J. C. Hendriks, Koen V. Hindriks, Catholijn M. Jonker. *Autonomous Agents and Multi-Agent Systems* (2016). <https://doi.org/10.1007/s10458-015-9309-1> Context: "The excerpt frames opponent modeling in bilateral negotiation around preference estimation, strategy prediction, and opponent classification. It distinguishes offline models, built from historical negotiations, from online models, learned during one negotiation under limited observations and possible real-time constraints. Regarding preference-model granularity, a model may estimate only offers likely to be accepted, such as through the opponent's reservation value; richer models can estimate acceptance probabilities, ordinal preference rankings, or a full cardinal preference profile. The stated motivations include minimizing negotiation costs, adapting to the opponent, and reaching win-win agreements. In particular, models may estimate the opponent's reservation value, deadline, and concessions: reservation value can help identify acceptable bids, deadline estimates can support concession elicitation, and concession estimates can enable reciprocation. However, this excerpt does not discuss the distinction between aleatoric and epistemic uncertainty, the value of perfect information about a counterpart's reservation value, or formal identifiability/confounding/impossibility results concerning reservation value, deadline, and concession rate from offer sequences. It only indicates that these quantities are targets of estimation and that online learning is difficult because observations are limited."

[3.9] Learning about the opponent in automated bilateral negotiation: a comprehensive survey of opponent modeling techniques. Tim Baarslag, Mark J. C. Hendriks, Koen V. Hindriks, Catholijn M. Jonker. *Autonomous Agents and Multi-Agent Systems* (2016). <https://doi.org/10.1007/s10458-015-9309-1> Context: "The excerpt does not explicitly distinguish aleatoric from epistemic uncertainty, nor does it discuss the value of perfect information about the counterpart's reservation value. It presents opponent modeling as learning several attributes, including acceptance strategy, deadline, preference profile, and bidding strategy. Reservation value estimation is based on behavioral assumptions—specifically, that concessions cease near the reservation value and often as the deadline approaches—and may use Bayesian learning, nonlinear regression, or related techniques. The passage indicates potential parameter confounding: under some models, reservation value and deadline are structurally related, so one may be derived from the other rather than independently identified. Other approaches infer tactic type and tactic parameters from observed offers, assuming a time-, behavior-, or resource-dependent strategy. Thus, the excerpt supports the conclusion that identifiability depends strongly on model assumptions, domain knowledge, publicly or privately known deadlines, and the amount and type of observed behavior. However, it does not state formal impossibility results, prove non-identifiability from offer sequences, or analyze whether reservation value, deadline, and concession rate can jointly be recovered. Those questions would require consulting later sections or the cited primary studies."

[4.1] Multi-Criteria Multi-Participant Automated Negotiation. F Eshragh. 2019. <https://ucalgary.scholaris.ca/server/api/core/bitstreams/b551ba20-5804-48a8-9750-4a889d0807f2/content> Context: "The excerpt directly supports the use of recursive Bayesian filtering and particle filtering for learning opponent preferences during a negotiation session. Because the proposer agent initially has no prior knowledge of participants and receives only feedback on offered proposals, it repeatedly updates its opponent model through prediction and correction phases. The approach uses unscented particle filtering to estimate opponents' criteria and recursively learn parameters of fuzzy preference models in real time. At each negotiation round, participant feedback and supporting arguments provide new evidence, while a probabilistic

graphical model uses the learned preference limits to construct subsequent proposals. This describes sequential Bayesian belief updating and progressive refinement of opponent-preference estimates as offers accumulate, although the excerpt does not explicitly use the terms “posterior concentration,” “posterior narrowing,” or “convergence.” It also does not mention a Gaussian process opponent model; the stated preference representation is based on fuzzy functions, and the filtering method is an unscented particle filter. The reported experiments in energy-system development and real-estate service found improved negotiation efficiency and agreement in fewer rounds, without requiring prior understanding of the participants."

[4.2] Multi-Criteria Multi-Participant Automated Negotiation. F Eshragh. 2019. <https://ucalgary.scholaris.ca/server/api/core/bitstreams/b551ba20-5804-48a8-9750-4a889d0807f2/content> Context: "The excerpt is relevant to Bayesian opponent-preference learning and particle-filter-style updating, but it does not specifically discuss Gaussian-process opponent models or prove posterior-concentration convergence. It frames opponent modeling as estimating a participant's preference profile, which determines how proposals are evaluated. Several existing approaches are reviewed: learning issue weights from concession behavior, classifying opponents into predefined preference classes, mining historical negotiation data, and using heuristic search when prior data or finite classes are unavailable. Bayesian methods update probabilities over preference hypotheses as proposals and other evidence accumulate during negotiation. The excerpt also notes that classification can suffer from scalability problems when utility models contain many attributes and continuous parameters, while historical-data methods may be unusable in new contexts. For the study's fuzzy utility models, each participant assigns fuzzy scores to proposal attributes, and proposal scores are obtained through defuzzification and weighted averaging. Since the unknown model contains more parameters than simple issue weights, the authors reject basic weight-learning and predefined-class approaches. Instead, they propose recursive Bayesian filtering to estimate fuzzy-model parameters from participant feedback received at each negotiation round. RBF is described as estimating an unknown system state from measurements over time, with Kalman and particle filtering identified as implementations. Thus, the passage supports sequential Bayesian belief updating and potential posterior narrowing as offers/feedback accumulate, but provides no formal convergence or concentration result."

[4.3] Multi-Criteria Multi-Participant Automated Negotiation. F Eshragh. 2019. <https://ucalgary.scholaris.ca/server/api/core/bitstreams/b551ba20-5804-48a8-9750-4a889d0807f2/content> Context: "The excerpt is relevant to Bayesian opponent modeling and sequential learning of opponent preferences, but it does not specifically describe a Gaussian process model, a particle-filter implementation, or a formal posterior-concentration/convergence result. It proposes recursive Bayesian filtering (RBF) to estimate the parameters of opponents' fuzzy utility functions. The model represents participants' preferences through fuzzy scores rather than binary acceptance or rejection thresholds. During each negotiation round, the proposer receives feedback consisting of the score assigned to an offered proposal and any natural-language arguments supporting the stakeholder's decision. The arguments are processed by a text-processing module before being supplied to the estimation module. The proposer uses this recursively learned information to approximate opponents' criteria and utility functions, enabling it to prepare proposals more likely to satisfy stakeholders and reduce the number of negotiation rounds. The stated assumptions include one proposal-making agent, logical opponents whose strategies do not change during negotiation, and a fixed set of negotiation issues. Thus, the passage supports an interpretation of Bayesian belief updating from accumulating offer evaluations, but it provides no explicit theorem or empirical claim about posterior narrowing, asymptotic consistency, or convergence of opponent beliefs over a session."

[5.1] Automated Negotiation with Gaussian Process-based Utility Models. Haralambie Leahu, Michael Kaisers, Tim Baarslag. Proceedings of the Twenty-Eighth International Joint Conference on Artificial Intelligence (2019). <https://doi.org/10.24963/ijcai.2019/60> Context: "The excerpt describes a Gaussian-process-based stochastic utility model for automated negotiation under incomplete information. The model represents uncertainty about both the user's and opponent's preferences through random utility functions, with the agent maintaining quantitative beliefs and updating them as new interaction data becomes available. The paper specifically emphasizes learning from comparative queries, in which the user compares pairs of alternatives, rather than providing numerical utility values. Gaussian processes provide a flexible framework for modeling random preference functions, and Bayesian posterior updating—implemented using a Laplace approximation—allows new observations to be incorporated through GP parameter updates. These ideas are integrated into a belief Markov Decision Process for sequential offer/counter-offer negotiation. However, the provided excerpt does not mention a particle filter, nor does it provide a formal result concerning posterior concentration or convergence of an opponent model during a negotiation session. It supports the general concept that beliefs about opponent preferences should become updated as offers or other interaction data accumulate, but it does not establish that the posterior necessarily narrows, quantify a convergence rate, or specify conditions under which convergence occurs. Thus, the excerpt is relevant to GP opponent modeling and Bayesian belief updating, but insufficient for claims about particle-filter learning or posterior convergence."

[5.2] Automated Negotiation with Gaussian Process-based Utility Models. Haralambie Leahu, Michael Kaisers, Tim Baarslag. Proceedings of the Twenty-Eighth International Joint Conference on Artificial Intelligence (2019). <https://doi.org/10.24963/ijcai.2019/60> Context: "The excerpt is relevant to Bayesian modeling of uncertainty about an opponent's preferences in automated bargaining, although it does not explicitly distinguish aleatory uncertainty from epistemic uncertainty, formulate adversarial risk analysis, or analyze the value of information about an opponent's reservation value. It proposes a stochastic utility model in which the agent represents the user's and opponent's utilities as random functions, with randomness reflecting uncertainty about preferences. These quantitative beliefs are updated as new information is obtained through interaction. Gaussian Processes provide a flexible framework for representing uncertain utility functions and incorporating additional observations through parameter updates. The paper emphasizes that preference information is often incomplete and that directly eliciting numerical utilities is costly, inconvenient, or inconsistent; instead, agents can learn ordinal preferences through comparative queries. This supports a Bayesian interpretation in which observations about offers, counteroffers, or user comparisons reduce epistemic uncertainty about utility rankings. The model is integrated into a belief Markov Decision Process for sequential negotiation, allowing uncertain beliefs to inform negotiation decisions. However, the excerpt provides no specific treatment of irreducible outcome randomness (aleatory uncertainty), no explicit decomposition between uncertainty types, and no direct calculation of the expected value of querying information about the opponent's reservation value."

[5.3] Automated Negotiation with Gaussian Process-based Utility Models. Haralambie Leahu, Michael Kaisers, Tim Baarslag. Proceedings of the Twenty-Eighth International Joint Conference on Artificial Intelligence (2019). <https://doi.org/10.24963/ijcai.2019/60> Context: "The excerpt provides a Bayesian utility-modeling framework relevant to beliefs about an opponent's preferences in automated negotiation, but it does not explicitly discuss adversarial risk analysis, aleatory versus epistemic uncertainty, or an opponent reservation value. It represents uncertainty about utility functions using probability laws, including Gaussian-process-based utility beliefs specified by a mean and covariance function. The agent maintains both a user utility model and an opponent model, updating them during interactions. Opponent behavior is represented through an acceptance probability under belief η , while the excerpt cautions that using opponent utility to formalize behavior generally requires additional modeling assumptions. The

framework evaluates actions through expected rewards: accepting an offer yields expected user utility, whereas making an offer incorporates the opponent's acceptance probability. Preference elicitation queries have discount or "bother" costs, and an optimal elicitation policy maximizes the expected immediate reward after updating beliefs. This supports a general value-of-information interpretation: information is worthwhile when the expected improvement in negotiation reward exceeds its elicitation cost. However, the specific value of information about an opponent's reservation value is not developed, and no explicit separation between randomness in opponent behavior (aleatory uncertainty) and uncertainty about the opponent's preferences or model (epistemic uncertainty) appears in the excerpt."

[6.1] Preference Estimation via Opponent Modeling in Multi-Agent Negotiation. Yuta Konishi, Kento Yamamoto, Eisuke Sonomoto, Rikuho Takeda, Ryo Furukawa, Yusuke Muraki, Takafumi Shimizu, Kazuma Fukumura, Yuya Kanemoto, Takayuki Ito, Shiyao Ding. ArXiv (2026). <https://doi.org/10.48550/arxiv.2604.15687> Context: "The excerpt presents a Bayesian opponent-modeling approach for estimating preferences during multi-party, multi-issue negotiation. Each opponent is represented through hypotheses about private issue-level score functions and reservation thresholds. As negotiation proceeds, the model processes both proposed deals and natural-language utterances. An LLM extracts qualitative preference signals from utterances, which are converted into probabilistic observations. The likelihood of these signals is modeled using Luce's Choice Axiom, while the likelihood of a proposed deal is approximated with a normal distribution. These likelihoods are combined to update the posterior probability of each hypothesis through Bayesian fusion, enabling dynamic belief tracking as evidence accumulates across rounds. The framework is intended to overcome the limitations of numerical-only opponent models and direct LLM inference, particularly in settings with incomplete information and qualitative conversational cues. The excerpt does not specifically describe a Gaussian-process opponent model, a particle filter, or a formal theorem establishing posterior concentration or convergence of beliefs during a negotiation session. It does, however, provide a related Bayesian mechanism in which successive offers and utterance-derived signals can narrow or revise posterior beliefs over possible opponent preferences."

[6.2] Preference Estimation via Opponent Modeling in Multi-Agent Negotiation. Yuta Konishi, Kento Yamamoto, Eisuke Sonomoto, Rikuho Takeda, Ryo Furukawa, Yusuke Muraki, Takafumi Shimizu, Kazuma Fukumura, Yuya Kanemoto, Takayuki Ito, Shiyao Ding. ArXiv (2026). <https://doi.org/10.48550/arxiv.2604.15687> Context: "The excerpt is relevant to Bayesian belief-state modeling for negotiation, but it does not specifically discuss the BOND model, distilling Bayesian beliefs into language models, Brier scores, or calibration metrics. It describes a Bayesian opponent-modeling framework in which an agent maintains a finite hypothesis space over possible opponent preferences. Each hypothesis consists of issue weights and issue-specific evaluation functions, producing an additive estimated utility for candidate deals. Observed numerical offers contribute a likelihood based on the proximity between estimated utility and an assumed concession-based target utility. Natural-language utterances are processed by an LLM into structured signals consisting of a target and stance, such as preferring or opposing an issue or option. Linguistic likelihoods are computed using Luce's Choice Axiom, with issue-preference probabilities proportional to the corresponding hypothesis weights. The framework combines numerical-offer and linguistic likelihoods through Bayesian updating, allowing multimodal observations to refine beliefs about opponent preferences during negotiation. The text assumes linguistic signals are broadly truthful, while noting that a reliability parameter could be introduced to reduce or adapt the influence of language based on consistency with observed offers and dialogue. These ideas could provide background for belief-state representation and uncertainty handling, but no explicit belief distillation procedure, BOND architecture, opponent-type prediction evaluation, Brier score, or calibration analysis appears in the excerpt."

[7.1] Distilling Bayesian Belief States into Language Models for Auditable Negotiation. Zongqi Cui, Baihan Lin. ArXiv (2605). <https://doi.org/10.48550/arxiv.2605.04507> Context: "The excerpt presents BOND (Bayesian Opponent-belief Negotiation Distillation), a framework for making negotiation agents' beliefs about opponent types explicit and auditable. In the CaSiNo setting, the hidden opponent type is one of six private priority orderings over Food, Water, and Firewood. An LLM-based Bayesian teacher evaluates each dialogue context against these six hypotheses, converts the scores into a posterior distribution, and uses that posterior for accept, reject, or counteroffer decisions. The teacher achieves a mean turn-level Brier score of 0.085 on 150 held-out dialogues and 1,054 predictions. An 8B student is trained to produce negotiation actions alongside normalized posterior beliefs in tagged text; it reaches Brier 0.114, improving substantially over the uniform six-ordering reference of approximately 0.139. A larger 70B structured-chain-of-thought baseline has better accept/reject F1 but much poorer posterior calibration, with Brier 0.194. The paper uses posterior trajectories, belief-versus-policy error decomposition, and posterior-prefix interventions to assess auditability. Its findings suggest that distillation preserves a scoreable and reasonably calibrated belief report, while the causal connection between corrected beliefs and selected actions remains comparatively weak."

[7.2] Distilling Bayesian Belief States into Language Models for Auditable Negotiation. Zongqi Cui, Baihan Lin. ArXiv (2605). <https://doi.org/10.48550/arxiv.2605.04507> Context: "The BOND framework (Bayesian belief-state distillation) separates negotiation into language interpretation, Bayesian belief revision, menu/action selection, and response generation. In the CaSiNo domain, the opponent's hidden type is represented as one of six possible orderings over FOOD, WATER, and FIREWOOD priorities. At each dialogue turn, a LoRA-adapted Llama-3.1-8B model scores how compatible the full dialogue history is with each ordering. These compatibility scores are clipped and temperature-transformed, then combined with the previous posterior under a Bayesian update, using a uniform prior unless otherwise specified. The resulting categorical posterior is used both for opponent-type prediction and for scoring candidate allocations through posterior expected opponent utility. The posterior is evaluated directly with turn-level Brier score, making calibration and belief quality measurable rather than relying only on negotiation outcomes. In the main teacher evaluation on 1,054 turn-level predictions from 150 held-out dialogues, the reported mean Brier score is 0.085; lower values indicate more accurate probabilistic predictions. The teacher's posterior, intent, selected content, and utterance are distilled into an 8B student that emits tagged structured text, allowing the model's belief state and action rationale to be inspected and audited at every turn. The excerpt does not provide additional calibration metrics such as reliability diagrams or expected calibration error."

[7.3] Distilling Bayesian Belief States into Language Models for Auditable Negotiation. Zongqi Cui, Baihan Lin. ArXiv (2605). <https://doi.org/10.48550/arxiv.2605.04507> Context: "BOND distills Bayesian belief states about an opponent's priority ordering into an 8B language model that emits posterior beliefs as tagged text alongside negotiation actions. The excerpt reports that the distilled student preserves much of the Bayesian teacher's posterior calibration, although standard supervised fine-tuning preserves belief reports more reliably than the causal connection from posterior beliefs to actions. In Protocol 3 on 150 held-out CaSiNo dialogues, the table reports Brier-related results for the Bayesian teacher and distilled student, with values shown as 0.048 and 0.190 for the teacher and 0.190 and 0.114 for the student across the displayed evaluation columns; the excerpt's formatting does not identify all column headers. Teacher and student Brier scores are computed from native single-pass posterior outputs, while the 70B baseline's Brier score is derived from majority-vote shares over five self-consistency samples. In a cross-domain Deal-or-No-Deal test, zero-shot renamed transfer achieves a DND-normalized Brier score of 0.1335, compared with 0.3190 for direct posterior prompting; few-shot adaptation improves this to 0.1166. These

DND scores use a different normalization from CaSiNo and should only be compared within their respective sections. The posterior is an auditable computational belief state rather than a direct measurement of human mental states."

[8.1] Deal, or no deal (or who knows)? Forecasting Uncertainty in Conversations using Large Language Models. Anthony B. Sicilia, Hyunwoo Kim, Khyathi Raghavi Chandu, Malihe Alikhani, Jack Hessel. ArXiv (2024). <https://doi.org/10.48550/arxiv.2402.03284> Context: "The excerpt is relevant to calibration and Brier-score-based evaluation of negotiation forecasting, but it does not mention the BOND model, Bayesian belief-state distillation, or opponent-type prediction specifically. The paper formulates conversation uncertainty modeling as forecasting binary outcomes from partial, multi-party dialogues. It evaluates whether predicted probabilities are calibrated: among conversations assigned probability p , the observed outcome frequency should also be p . However, calibration alone may be insufficient because a constant forecast equal to the overall outcome mean can be calibrated without capturing conversation-specific uncertainty. The authors therefore emphasize strictly proper scoring rules, which jointly encourage calibration and accurate conditional probabilities. Their main tangible metric is the Brier score, defined as the mean squared error between forecast probabilities and realized outcomes. They also use the Brier skill score, $1 - \text{BS}/\text{BS}_{\text{ref}}$, where the reference may be the constant prior prediction $E[\text{O}]$. This skill score can be interpreted as the percentage of outcome variance explained by the forecaster or improvement over prior knowledge. The paper considers both implicit forecasts derived from model token probabilities and direct natural-language probability forecasts, and reports experiments on deals, decisions, and emotional conflicts in negotiation-like settings."

[8.2] Deal, or no deal (or who knows)? Forecasting Uncertainty in Conversations using Large Language Models. Anthony B. Sicilia, Hyunwoo Kim, Khyathi Raghavi Chandu, Malihe Alikhani, Jack Hessel. ArXiv (2024). <https://doi.org/10.48550/arxiv.2402.03284> Context: "The excerpt is relevant to calibration in negotiation forecasting, but it does not mention distilling Bayesian belief states, a BOND model, opponent-type prediction, or the Brier score specifically. It frames negotiation forecasting as predicting uncertainty about binary conversation outcomes, such as whether a deal occurs, rather than merely selecting the most likely next utterance. A forecaster maps a partial dialogue to a probability estimate for the outcome. Calibration is defined using the condition that, among conversations assigned probability p , the empirical mean outcome should equal p . The paper proposes FortUne Dial, an expansion of conversation forecasting covering eight negotiation-related datasets and tasks involving deals, decisions, and emotional conflicts. Models can represent uncertainty implicitly through token-score distributions or explicitly by generating probability statements. The study evaluates calibration with uncertainty-aware metrics, enabling abstention on individual instances when the model is highly uncertain, and investigates GPT-4, Llama-2, and Zephyr-style models. It also proposes supervised and off-policy reinforcement-learning fine-tuning strategies intended to improve calibration, including for smaller open-source models. Thus, the passage provides conceptual background for calibrated probabilistic negotiation forecasting, but no evidence about Bayesian belief-state distillation, BOND, opponent-type classification, or Brier-score results."

[9.1] Adversarial risk analysis: An overview. David Banks, Víctor Gallego, Roi Naveiro, David Ríos Insua. Wiley Interdisciplinary Reviews: Computational Statistics (2022). <https://doi.org/10.1002/wics.1530> Context: "The excerpt presents adversarial risk analysis (ARA) as a Bayesian, decision-theoretic framework for reasoning about intelligent opponents under uncertainty. Rather than assuming a Nash equilibrium or perfectly rational adversary, the analyst constructs a model of the opponent's goals, resources, beliefs, capabilities, and strategic reasoning, placing subjective probability distributions over quantities that are unknown. Solving the decision problem from the opponent's perspective produces a probability distribution over the opponent's possible actions, which the analyst uses to maximize expected utility. This

framework is directly relevant to bargaining and negotiation because it supports modeling uncertainty about an opponent's preferences and likely responses to alternative proposals. The excerpt distinguishes aleatory uncertainty—uncertainty in outcomes conditional on the opponents' choices—from epistemic uncertainty, which concerns imperfect knowledge of the opponent's utility function, beliefs about outcome probabilities, and resources. An opponent's reservation value would therefore be an example of an uncertain preference-related quantity that could be represented probabilistically. However, the supplied pages do not specifically discuss bargaining, reservation values, or the value of information. They establish the Bayesian modeling foundation for assessing whether additional information about an opponent could improve expected utility, but no explicit value-of-information formula or application is provided here."

[9.2] Adversarial risk analysis: An overview. David Banks, Víctor Gallego, Roi Naveiro, David Ríos Insua. Wiley Interdisciplinary Reviews: Computational Statistics (2022). <https://doi.org/10.1002/wics.1530> Context: "The excerpt provides a useful ARA distinction for analyzing an opponent in bargaining or negotiation. Aleatory uncertainty concerns randomness in the outcome conditional on the parties' decisions; in the auction illustration, it includes uncertainty about the book's condition and the value realized after bids are made. Epistemic uncertainty concerns what the analyst does not know about the opponent, including the opponent's utility function or valuation, beliefs about outcome probabilities, and available resources. A Bayesian ARA model represents these unknowns with subjective probability distributions, potentially obtained through elicitation. In the auction example, Daphne assigns distributions to Apollo's valuation and to the fraction of that valuation he may bid, then integrates over these uncertainties to obtain a subjective probability of winning and optimize expected utility. This framework can analogously represent uncertainty about an opponent's reservation value in negotiation and support decisions about whether information collection is worthwhile. However, the excerpt does not explicitly define or calculate the value of information, nor does it discuss bargaining-specific reservation values. It does emphasize that analysts must distinguish uncertainty about the opponent's attributes from concept uncertainty—uncertainty about how the opponent makes decisions—and can model nonstandard or boundedly rational behavior."

[10.1] Uncertainty-Aware Opponent Modeling for Deep Reinforcement Learning. Likun Yang, Pei Xu, Shiyue Cao, Yongjian Ren, Xiaotang Chen, Kaiqi Huang. ArXiv (2025). <https://doi.org/10.5555/3709347.3743861> Context: "The excerpt describes UAOM (Uncertainty-Aware Opponent Modeling), a framework for deep reinforcement learning that explicitly addresses two uncertainty sources in opponent modeling. Aleatoric uncertainty is attributed to inherent randomness in opponent behavior, while epistemic uncertainty reflects incomplete knowledge of the system dynamics. UAOM uses a hybrid behavior modeling module that combines deterministic and probabilistic encoders and decoders, rather than relying only on deterministic representations. The probabilistic component models uncertainty through distributional latent representations, including mean and variance and reparameterized sampling, while the deterministic component complements this representation. The framework also introduces an opponent-aware dynamic modeling module to capture transition information influenced by opponents, using the controlled agent's action and subsequent environmental feedback. An encoder-decoder structure, recurrent encoding, and state/reward reconstruction are used to model these dynamics. During execution, the learned encoder infers a current belief from partial observations and self-action history. Training jointly optimizes behavior modeling, dynamic modeling, latent-distribution regularization, and downstream reinforcement-learning losses; the paper reports using A2C for the downstream learner. This joint optimization is intended to reduce uncertainty, improve predictive accuracy, and support more effective reinforcement-learning decisions."

[10.2] Uncertainty-Aware Opponent Modeling for Deep Reinforcement Learning. Likun Yang, Pei Xu, Shiyue Cao, Yongjian Ren, Xiaotang Chen, Kaiqi Huang. ArXiv (2025). <https://doi.org/10.5555/3709347.3743861> Context: "The excerpt presents Uncertainty-Aware Opponent Modeling (UAOM) for deep reinforcement learning, targeting opponents whose behavior is highly stochastic. It distinguishes two forms of uncertainty in opponent modeling: aleatoric uncertainty, caused by intrinsic randomness in an opponent's actions, and epistemic uncertainty, caused by the difficulty of inferring complex behavioral patterns. UAOM addresses these uncertainties with a hybrid behavior-modeling module that combines probabilistic and deterministic networks in an ensemble. The probabilistic network represents the opponent's behavior as a distribution and captures aleatoric randomness, rather than producing only a single deterministic action prediction. The ensemble of models improves epistemic capacity by representing uncertainty associated with limited or incomplete knowledge of complicated opponent behaviors. UAOM also addresses uncertainty introduced when the opponent model is used by the downstream reinforcement-learning policy. Its opponent-aware dynamic modeling approach learns environmental dynamics in the presence of opponents, reducing the impact of uncertain opponent-model predictions on the controlled agent's Markov dynamics and learning process. The reported experiments cover simulated opponents with varying randomness in cooperative, competitive, and mixed-motive games, as well as human-agent interaction tasks. Results reportedly show particularly strong improvements over baselines when opponent randomness is high."

[10.3] Uncertainty-Aware Opponent Modeling for Deep Reinforcement Learning. Likun Yang, Pei Xu, Shiyue Cao, Yongjian Ren, Xiaotang Chen, Kaiqi Huang. ArXiv (2025). <https://doi.org/10.5555/3709347.3743861> Context: "The excerpt presents Uncertainty-Aware Opponent Modeling (UAOM), a framework for deep reinforcement learning that explicitly addresses two forms of uncertainty in opponent modeling. Aleatoric uncertainty arises from intrinsic randomness or high stochasticity in an opponent's behavior, while epistemic uncertainty concerns limitations in representing complex and diverse behavioral strategies. UAOM uses a Hybrid Behaviour Modeling Module combining deterministic and probabilistic networks. Both networks share a recurrent encoder that processes the controlled and modeled agents' history trajectories into an opponent behavioral embedding. The deterministic decoder predicts the opponent's exact observation and action trajectories, supporting the representation of structured and complex behavior. The probabilistic decoder models observations with a Gaussian distribution and actions with a Gumbel distribution, using reparameterized sampling to preserve trainability. This probabilistic component captures behavioral randomness and predicts distributions rather than single deterministic actions; observation outputs are trained with mean squared error, while action distributions use negative log-likelihood. The ensemble is intended to improve cognitive representation and mitigate epistemic uncertainty without losing the ability to model aleatoric variability. UAOM also adds opponent-aware dynamic modeling so that transition dynamics affected by opponent actions can provide more consistent features for downstream reinforcement learning."

[11.1] Evaluating the quality of opponent models in automated bilateral negotiations. M Hendrikx. 2012. https://repository.tudelft.nl/file/File_67920399-8fc9-48c5-b696-465f800d2fd5 Context: "The excerpt does not explicitly use the distinction between aleatoric and epistemic uncertainty. It frames uncertainty primarily as incomplete knowledge of the opponent's private preference profile and as possible model inaccuracy, while also emphasizing computational and protocol-related effects. The paper proposes comparing negotiation performance with and without opponent models, including a benchmark based on perfect knowledge of the opponent's preferences. This is relevant to the value of information, but the excerpt does not state that it estimates the value of perfect information specifically about the counterpart's reservation value; the benchmark concerns the preference profile more generally. The provided pages also do not present identifiability, confounding, or impossibility results concerning reservation value, deadline,

or concession rate inferred from offer sequences. They do establish that deadline type matters: time-based deadlines create a time/exploration trade-off because model computation reduces the number of bids that can be explored, whereas the study also considers round-based protocols. More broadly, model performance is said to depend on the negotiation setting, and a poor model may recommend unattractive bids or perform worse than using no model. Thus, these pages motivate questions about information value and inference limits but do not answer the specific reservation-value identifiability question."

[12.1] Evaluating the quality of opponent models in automated bilateral negotiations. M Hendrikx. 2012. https://repository.tudelft.nl/file/File_67920399-8fc9-48c5-b696-465f800d2fd5 Context: "The excerpt discusses uncertainty in opponent modeling primarily as incomplete or imperfect knowledge of the opponent's preference profile, rather than using the terminology of aleatoric versus epistemic uncertainty. It explicitly proposes comparing negotiation performance under perfect knowledge of the opponent's preferences with performance when that information is ignored, and separately comparing online modeling with no model. However, the stated perfect-knowledge comparison concerns the opponent's full preference profile, not specifically the value of information about the counterpart's reservation value. The excerpt emphasizes that model quality can be poor, that time-based learning creates a time/exploration trade-off, and that performance depends on the negotiation setting. It describes private preferences, alternating offers, and time or round-based protocols, but does not present identifiability results for reservation value, deadline, or concession rate from an offer sequence. Nor does it establish whether those quantities can be uniquely recovered, or discuss confounding and impossibility theorems. The supplied passage therefore provides conceptual motivation for studying value of opponent information and inference limitations, but does not document a formal aleatoric/epistemic distinction, a reservation-value value-of-information analysis, or specific identifiability conclusions."

[12.2] Evaluating the quality of opponent models in automated bilateral negotiations. M Hendrikx. 2012. https://repository.tudelft.nl/file/File_67920399-8fc9-48c5-b696-465f800d2fd5 Context: "The excerpt presents evaluation as an outcome-based benchmarking methodology for online opponent models embedded in complete negotiation agents. It emphasizes that model performance depends strongly on the negotiation setting, including the bidding strategy using the model, the opponent strategy, and scenario characteristics. For fair comparisons, non-model components—especially the bidding and acceptance strategies—should be fixed. The experimental agents use a variant of a time-dependent tactic, generate similarly preferred bids, and use the opponent model to select the bid that is best for the opponent. Models are embedded in four agents with different concession parameters ($e = 0.1, 0.2, 1.0, \text{ and } 2.0$), allowing the effect of concession speed to be examined. The benchmark includes eight opponent strategies: three cooperative/high-concession agents and five competitive agents, including ANAC agents. It also uses seven negotiation scenarios selected to span domain size, bid distribution, and opposition. The excerpt does not mention proper probabilistic scoring rules such as log score, Brier score, or CRPS, nor does it describe calibration diagnostics. It likewise does not report specific point-error metrics such as RMSE for reservation value or utility distance. Instead, the stated evaluation focuses on performance gains and negotiation outcomes when models guide bid selection, assessed across varied agents, opponents, and domains."

[12.3] Evaluating the quality of opponent models in automated bilateral negotiations. M Hendrikx. 2012. https://repository.tudelft.nl/file/File_67920399-8fc9-48c5-b696-465f800d2fd5 Context: "The excerpt describes an evaluation framework centered on the effect of opponent modeling on negotiation performance, rather than specifying detailed probabilistic scoring or calibration diagnostics. Models are evaluated in both time-based and round-based bilateral alternating-offers negotiations, across various negotiation settings. The techniques are isolated from complete state-of-the-art negotiation strategies so that model performance can be compared directly. The stated benchmarks include ignoring the opponent's